WORKED SOLUTION

STEP III 1987, Question 1

Pure • Divisibility of Factorials

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q1

The key idea

The factorial $(n-1)! = 1 \times 2 \times \cdots \times (n-1)$ contains every whole number smaller than n. If n can be built as a product of two *different* numbers smaller than n, both appear in the factorial and n divides it. So the question is really about which n *can't* be split up like that: primes clearly can't, and squares of primes need a closer look. Trying the first dozen values by hand points straight at the answer.

Solution

Experiment first

A quick table is worth making:

n	1	2	3	4	5	6	7	8
$(n-1)!$	1	1	2	6	24	120	720	5040
$n \mid (n-1)!$?	yes	no	no	no	no	yes	no	yes

The failures are $2, 3, 4, 5, 7$: the primes so far, together with 4. We now prove that this pattern continues.

Claim. n does not divide $(n-1)!$ exactly when n is prime or $n = 4$.

$n = 1$

Here $(n-1)! = 0! = 1$, and 1 divides 1. So $n = 1$ is *not* in the set.

$n = p$ **prime**

Each factor $1, 2, \ldots, p-1$ of $(p-1)!$ is smaller than p and positive, so none is divisible by p. Since p is prime, if p divided the product it would have to divide one of the factors (Euclid's lemma). Hence $p \nmid (p-1)!$, and every prime is in the set.

n composite

Let p be the smallest prime factor of n and write $n = pm$. Because n is composite, $m > 1$, and every prime factor of m is at least p, so $m \geq p$. There are two cases.

Case 1: $m > p$. Then $1 < p < m < n$, so p and m are two *different* numbers in the list $1, 2, \ldots, n-1$. Their product $pm = n$ therefore divides $(n-1)!$.

Case 2: $m = p$, **so** $n = p^2$.

> **Ask yourself** Why doesn't Case 1's argument work here, and what can replace it?

The factor p appears only once in the list $1, \ldots, n-1$, so we can't use it twice. Instead use p and $2p$. If $p \geq 3$ then

$$2p < p \cdot p = n,$$

so p and $2p$ are two different numbers in the list $1, 2, \ldots, n-1$. Hence their product $2p^2 = 2n$ divides $(n-1)!$, and so does n.

If $p = 2$ this fails, because $2p = 4 = n$ is not in the list. Indeed $n = 4$ gives $3! = 6$, and $4 \nmid 6$. So 4 is in the set.

Conclusion

Every composite n other than 4 divides $(n-1)!$, and so does $n = 1$. Therefore the required set is

$$\{\, n : n \text{ is prime} \,\} \cup \{4\} = \{2, 3, 4, 5, 7, 11, 13, 17, \ldots\}.$$

Checks

The table above agrees with the claim for $n \leq 8$. Also try a prime square: $n = 9$ gives $8! = 40320 = 9 \times 4480$, as Case 2 predicts (using the factors 3 and 6). A computer check of every $n \leq 2000$, computing $(n-1)!$ mod n directly, finds no exceptions to the claim.

Common slips

- Arguing "write $n = ab$ with $a, b < n$, so $ab \mid (n-1)!$" without insisting that $a \neq b$. That argument wrongly "proves" that $4 \mid 3!$.
- Forgetting $n = 1$. Since $0! = 1$, it is not in the set.
- Saying a prime p fails to divide $(p-1)!$ "because p isn't one of the factors". The real reason is that p is prime and divides none of the factors; a composite number can divide a product without being one of the factors.

Beyond STEP

There is a sharper fact for primes, called Wilson's theorem: $(p-1)! \equiv -1 \pmod{p}$. Together with this question, it gives a (very slow) test for primality: $n > 1$ is prime exactly when $(n-1)! \equiv -1 \pmod{n}$. Wilson's theorem is proved by pairing each number with its inverse modulo p, which is a first taste of the group structure of the integers modulo p. Davenport's *The Higher Arithmetic* covers this beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.